MAT 221 - CALCULUS I
Lesson 17: - Arc Length

DR. M. KOORAPETSE and DR. I. Humbu

August 26, 2026

University of Botswana
Department of Mathematics

Arc Length

- If f' is continuous on $[a, b]$, then the **length (arc length)** of the curve $y = f(x)$ from the point $A = (a, f(a))$ to the point $B = (b, f(b))$ is the value of the integral

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \tag{1}$$

- If g' is continuous on $[c, d]$, then the **length (arc length)** of the curve $x = g(y)$ from the point $A = (g(c), c)$ to the point $B = (g(d), d)$ is the value of the integral

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \tag{2}$$

- If the curve is **parametric**, that is $y = g(t)$ and $x = f(t)$ between $t_1 \leqslant t \leqslant t_2$ length of curve is

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \tag{3}$$

Example 1

Determine the length of $y = \ln(\sec x)$ between $0 \leqslant x \leqslant \frac{\pi}{4}$.

Solution

Lets get the derivative

$$\frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x \quad \Rightarrow \quad \left(\frac{dy}{dx}\right)^2 = \tan^2 x$$

Also lets get the root out of the way

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = |\sec x| = \sec x$$

Note that we could drop the absolute value bars here since secant is positive in the range given. The arc length is then,

$$\begin{aligned} L &= \int_0^{\frac{\pi}{4}} \sec x \, dx \\ &= \ln|\sec x + \tan x||_0^{\frac{\pi}{4}} \\ &= \ln(\sqrt{2} + 1). \end{aligned}$$

Example 2

Determine the length of $x = \frac{2}{3}(y-1)^{\frac{3}{2}}$ between $1 \leqslant y \leqslant 4$.

Solution

$$\frac{dx}{dy} = (y-1)^{\frac{1}{2}} \quad \Rightarrow \quad \left(\frac{dx}{dy}\right)^2 = ((y-1)^{\frac{1}{2}})^2 = y-1$$

also

$$\sqrt{1+\left(\frac{dx}{dy}\right)^2} = \sqrt{1+y-1} = \sqrt{y} = y^{\frac{1}{2}}$$

The arc length is then,

$$\begin{aligned} L &= \int_1^4 y^{\frac{1}{2}}\, dy \\ &= \frac{2}{3} y^{\frac{3}{2}} \Big|_1^4 \\ &= \frac{14}{3}. \end{aligned}$$

Example 3

Find the length of the curve $x = 5(2t - \sin 2t)$, $y = 10\sin^2 t$ between $t = 0$ and $t = \pi$.

Solution

$$\frac{dx}{dt} = 5(2 - 2\cos 2t) = 10 - 10\sin 2t$$

and

$$\frac{dy}{dt} = 20\sin t\cos t = 10\sin 2t$$

We have

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 100 + 100\cos^2 2t - 200\cos t + 100\sin^2 2t \\ &= 100 + 100(\cos^2 2t + \sin^2 2t) - 200\cos 2t \\ &= 200 - 200\cos 2t \\ &= 200(1 - \cos 2t) \\ &= 400\sin^2 t, \qquad \because \cos 2t = 1 - 2\sin^2 t\end{aligned}$$

Therefore $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 20\sin t.$

Then the length of arc is

$$\begin{aligned} L &= \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \\ &= \int_0^{\pi} 20 \sin t \, dt \\ &= 20(-\cos t)\big|_0^{\pi} \\ &= 40. \end{aligned}$$